

Executive Summary

This project develops a Credit Risk Intelligence Platform that predicts loan default risk and evaluates portfolio resilience under changing economic conditions.

Using over 1.3 million Lending Club loans and macroeconomic indicators from the Federal Reserve, machine learning models were developed to estimate default probability and assess risk under economic stress scenarios.

Key Findings:

- Overall default rate: 19.98%
- Logistic Regression ROC-AUC: 0.648
- Random Forest ROC-AUC: 0.662
- FICO score was the strongest protective factor.
- Loan amount and debt-to-income ratio were the strongest risk factors.
- Economic stress testing demonstrated how portfolio risk can be evaluated under adverse economic conditions.